

Mesoscale Arrest without Progressive Slowing in a Protein Polymer with Reversible β -Sheet-like Order

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Abstract

Dynamic arrest is commonly pictured as a progressive slowing of motion. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) of the sequence-defined elastin-like polypeptide (VPAVG)₃₀ at 30 °C resolves a different route. Over 84–131 min at that temperature, the fast-decay amplitude of a two-step correlation function falls from 0.71 to 0.01 across 76–167 nm, while the fast relaxation time shows no systematic drift and keeps a near-diffusive Q dependence. The mobile population therefore shrinks without

9 slowing, and the largest length scales arrest first. Temperature-dependent SAXS/WAXS
10 resolves a reversible transition to a hierarchical assembly carrying β -sheet-like peaks
11 at $Q = 1.38$ and $0.72 \text{ \AA}^{-1}$ that vanish on cooling. Arrest by local conversion rather
12 than bulk stiffening implies that a partially arrested protein network retains fully
13 mobile regions, which bears on solute transport and on the maturation of biomolecular
14 condensates.

15 Amyloid-like β -sheet ordering occupies an unusual position in protein science and polymer
16 materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence
17 homology are associated with Alzheimer’s disease,¹ Parkinson’s disease,² type II diabetes,³
18 prion diseases,⁴ and systemic light-chain amyloidosis.⁵ Many amyloid fibrils are kinetically
19 persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-
20 zipper interfaces.⁶ Yet amyloid-like order is not intrinsically pathological: functional amyloids
21 store mammalian peptide hormones in pituitary secretory granules⁷ and build the extracellular
22 matrix of bacterial biofilms,⁸ and related assemblies are increasingly used as mechanically
23 robust biomaterials.⁹ What differs across these systems is not whether β -sheet-like order
24 forms but how long it lasts: hormone granules disassemble to release functional monomer,
25 whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a
26 wide range of stabilities rather than committing a sequence to one permanent structure,¹⁰
27 and where a sequence falls in that range is a materials-design variable.

28 Reversible β -sheet-like contacts are attractive as candidate physical junctions because
29 backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or
30 weakly packed interfaces can remain dynamic or reversible.^{10–13} Elastin-like polypeptides
31 (ELPs) are well suited to test this possibility because their sequence, molecular weight, and
32 lower critical solution temperature (LCST) behavior can be programmed recombinantly.^{14–18}
33 Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or
34 hierarchical assemblies depending on sequence, concentration, and thermal history.^{17–19} In
35 particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.^{20,21} Those studies establish the arrested state from time-averaged scattering and rheology. Neither can say how the assembly reaches it, and that is the question addressed here.

Whether an assembled state is still evolving is a question for a dynamical measurement. In colloidal gels, microscopic relaxation can continue to evolve after the time-averaged structure changes only weakly, a nonequilibrium process commonly termed physical aging.^{22,23} Related heterogeneous relaxation has been resolved during protein liquid–liquid phase separation,²⁴ protein gelation,²⁵ and cage relaxation in crowded protein solutions.²⁶ During arrested phase separation, a protein-rich morphology can therefore continue to reorganize through junction exchange, stress relaxation, and residual coarsening even when its SAXS profile changes modestly. A dynamical measurement can also separate two routes to arrest. If the assembly stiffens as a whole, relaxation times lengthen continuously. If it instead converts region by region, the material that is still mobile keeps its original relaxation rate while its share of the scattering falls. A bulk measurement averages the two together. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) probes the evolution directly through speckle decorrelation at the same low- Q length scales sampled by SAXS.

Here, temperature-dependent SAXS/WAXS and isothermal SA-XPCS provide complementary structural and dynamical views of concentrated aqueous (VPAVG)₃₀, an ELP built from 30 repeats of the pentapeptide valine-proline-alanine-valine-glycine. At 10 °C the scattering is dominated by molecular dimensions with no detectable mesoscale assembly; heating to 30 °C produces hierarchical low- Q scattering and high- Q peaks compatible with β -sheet-like packing, and cooling restores the low-temperature profile. In a separate experiment on a freshly loaded sample, isothermal SA-XPCS at 30 °C reveals a two-step decay whose fast fraction falls progressively with elapsed time. The mobile fraction therefore falls without the surviving dynamics slowing, while the local order that accompanies the assembly stays thermally reversible.

Figure 1 summarizes the SA-XPCS experiment. Its upper-left cartoon illustrates one

candidate pathway, not a direct observation: protein-poor solvent regions are hypothesized to appear and grow within the slower protein-rich phase, displacing and compressing it toward a coarsened, network-like morphology. This dynamic-asymmetry picture follows Tanaka’s viscoelastic phase-separation framework and is consistent with prior scattering studies of arrested ELP gels.^{20,27} The (VPAVG)₃₀ sample was approximately 100 mg mL⁻¹ in the buffer given below. Each 2 s SA-XPCS acquisition at Advanced Photon Source (APS) beamline 8-ID-I (Experimental Section) used a previously unexposed sample position to limit radiation damage (SI Section 6); temporal and azimuthal averaging yielded $I(Q)$ and multi-tau correlation analysis yielded $g_2(Q, \Delta t)$.

Figure 2 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were sealed in 2 mm quartz capillaries and measured at APS 12-ID-B using simultaneous SAXS/WAXS detection at 13.3 keV.^{28–30} The aliquot labeled “Measurement” had undergone more than 10 heating/cooling cycles, whereas the “Reference” aliquot had remained at 4 °C. At 10 °C, the two profiles overlap. A Guinier fit to the Reference data gives an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S3), consistent with a dispersed, unassembled low-temperature state lacking detectable mesoscale aggregates. The 10 °C profiles reach down to $Q = 0.0081$ Å⁻¹, so the absence of mesoscale assembly there is established for structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å⁻¹). Both aliquots also develop the same pair of high- Q features, at $Q = 1.38$ and 0.72 Å⁻¹ in the Reference aliquot and 1.36 and 0.74 Å⁻¹ in the Measurement aliquot. The Measurement aliquot alone makes the point for the high- Q features: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at 1 °C min⁻¹ (Figure S4). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on



Figure 1: SA-XPCS measurement of thermally induced $(\text{VPAVG})_{30}$ assembly at APS 8-ID-I. Each 100,000-frame set of sparse $20\ \mu\text{s}$ frames feeds two analyses. Averaging them in time gives the SAXS intensity $I(Q)$: how much structure the sample has at each length scale $2\pi/Q$, which grows at low Q on heating. Correlating them in time gives $g_2(Q, \Delta t)$: how fast the structure at that same length scale rearranges. A decay in g_2 means the density fluctuations at $2\pi/Q$ are still moving; a flat g_2 means they are frozen over the accessible delay times Δt . Both are sketched beside the detector (red: assembled; blue: dispersed; illustrative, not data). Center: the aluminum cell, liquid chamber 3 mm across. The photos, outlined in blue and red, show the ELP sample below and above the transition. Lower left: one $(\text{VPAVG})_{30}$ repeat, residues colored as keyed. Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; the graininess is sparse counting statistics, not resolved speckle. Upper left, stages i–iv: a conceptual illustration of the viscoelastic phase-separation pathway,^{20,27} in which protein-poor solvent pools (cyan) nucleate and grow within the still-continuous protein-rich phase (deepening red: more protein), coarsening it toward a network-like morphology.

every heating and returned to the same low-temperature state on every cooling (SI Section 7).
 Video S1 independently shows recovery of optical transparency after cooling.

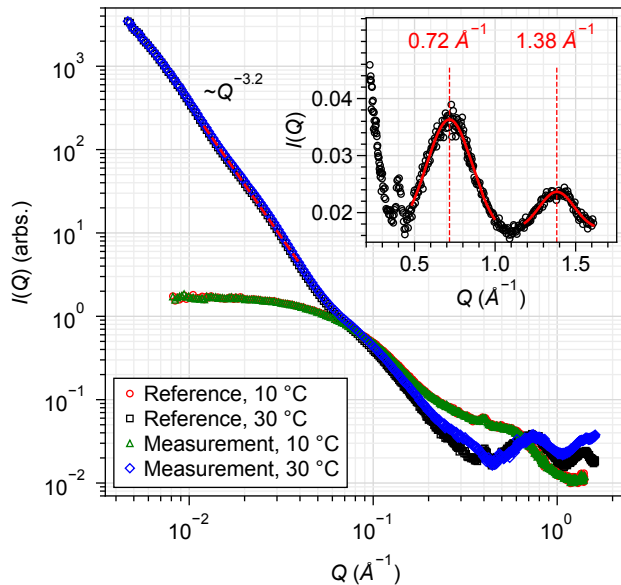


Figure 2: Temperature-dependent structure and reversibility of (VPAVG)₃₀. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50$ Å (Figure S3). Heating produces an approximate $Q^{-3.2}$ low- Q response and peaks at $Q = 1.38$ and 0.72 Å⁻¹, corresponding to real-space spacings of approximately 4.5 and 8.8 Å, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040$ Å⁻¹. Inset: the high- Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å⁻¹, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle scattering signifies density heterogeneity extending across the probed interval rather than a single resolved structural scale.³¹ Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.^{20,21} The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross- β -like assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.^{6,32} Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 Å⁻¹ (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å),

so the ordering extends over a few repeat spacings rather than forming extended cross- β crystallites. Their concurrent appearance with the low- Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β -sheet-like associations that are plausible, rather than uniquely established, physical junctions.

To resolve mesoscale arrest at fixed thermal driving force, a freshly loaded sample was held at 6 °C for more than 30 min, warmed to 20 °C at 5 °C min⁻¹, then to 30 °C at 1 °C min⁻¹ to limit overshoot. In the seven-cycle experiment the sample is still mobile at 31.9 °C and is arrested within a single ramp at 33.8 °C (Figure S4c), whereas at 30 °C the fast fraction at the largest length scale reaches zero only after about 130 min (Figure 3c). Temperature therefore sets the rate of network formation. Arrest itself is a kinetic crossover reached sooner the higher the temperature, as in colloidal gels, where the time to a given stage of arrest changes by about two orders of magnitude over a 1.5 K change in quench depth.³³ The waiting time t_w was defined from the moment the calibrated resistance-temperature detector (RTD) of the stage zone holding the cell read 30 °C. The sample was then followed by simultaneous SAXS and SA-XPCS for up to 7863 s (Figure 3). Each reported t_w labels a group of averaged consecutive acquisitions, each at a previously unexposed position, and is the acquisition time of the first member (SI Section 2.2). The SAXS intensity initially lies close to the 6 °C reference profile and then rises by more than two orders of magnitude at low Q as t_w advances, while remaining broad and monotonic (Figure 3a and Figure S5). That 6 °C reference retains a weak low- Q rise of its own, attributed to incompletely dissolved material in this loading (SI Section 7.5). This evolution shows increasing mesoscale density contrast without selection of a narrow domain spacing. Both the elapsed-time range and the Q range analyzed below are set by photon statistics rather than by a change in the sample’s dynamics. Before $t_w = 5040$ s the low- Q coherent intensity is too weak for a two-step correlation function to be resolved above the noise. At 31.9 °C the correlation decays essentially to its baseline (Figure S4c), so no second step is present there to resolve. Above $Q = 0.00827 \text{ \AA}^{-1}$ the scattered intensity has

fallen by more than an order of magnitude (Figure 3a) and the corresponding g_2 is dominated by counting noise. Reliable correlation functions are therefore restricted to $t_w \geq 5040$ s and the five lowest Q bins.

The intensity autocorrelation function is

$$g_2(Q, \Delta t) = \frac{\langle I(Q, t)I(Q, t + \Delta t) \rangle_t}{\langle I(Q, t) \rangle_t^2}, \quad (1)$$

where $I(Q, t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/Q$. At $Q = 0.00376 \text{ \AA}^{-1}$ (approximately 167 nm), the correlation functions show a two-step decay: an initial millisecond-scale relaxation followed by an apparent plateau that persists throughout the 2 s acquisition (Figure 3b). Resolving that first decay sets the frame rate. With τ_{fast} between 0.1 and 1.6 ms, a kilohertz acquisition would place the entire fast mode at or below its shortest delay and return only the slow component; the 20 μs frames used here sample it directly. With increasing t_w , the amplitude of the first decay decreases and the plateau rises. Such two-step relaxation and growth of a slow component are characteristic of heterogeneous kinetic arrest in protein and colloidal gels.^{22,25,34}

The correlation functions were parameterized by

$$g_2(Q, \Delta t) = 1 + \beta \left[f(Q) \exp \left[- \left(\frac{\Delta t}{\tau_{\text{fast}}(Q)} \right)^{p_{\text{fast}}} \right] + [1 - f(Q)] \exp \left[- \left(\frac{\Delta t}{\tau_{\text{slow}}(Q)} \right)^{p_{\text{slow}}} \right] \right]^2, \quad (2)$$

where β is the instrumental speckle contrast, measured independently on a static reference and held fixed (Figure S7), $f(Q)$ is the fitted fraction of fast dynamics, $1 - f(Q)$ is the fitted fraction of slow dynamics, and $\tau_{\text{fast}}(Q)$ and $\tau_{\text{slow}}(Q)$ are effective relaxation times. The stretching exponents carry no Q argument because they are shared across the five bins of a global fit. p_{fast} is shared further, across every elapsed time as well: a Kohlrausch τ is comparable between fits only when they use the same exponent, since the mean relaxation time carries a factor $\Gamma(1/p)/p$.³⁵ One value serves the whole series, $p_{\text{fast}} = 0.624 \pm 0.023$,

and the fit is insensitive to it: χ^2 per degree of freedom varies by under 1% between 0.55 and 0.70. p_{slow} is left free at each elapsed time. Relaxation times below are the mean times $\langle\tau\rangle = \tau \Gamma(1/p)/p$, not the Kohlrausch τ itself. Fitted this way, the two-mode form describes the observed first decay and persistent plateau without assigning either mode to a unique molecular mechanism. Because τ_{slow} frequently exceeds the acquisition window, it is an extrapolated descriptor rather than a directly sampled terminal time; the primary quantitative descriptor is the directly constrained fast fraction f .

The fitted fast fraction decreases monotonically with elapsed time at every measured Q . At fixed $t_w = 5040$ s, f increases with Q , from 0.42 at $Q = 0.00376 \text{ \AA}^{-1}$ (167 nm) to 0.71 at $Q = 0.00827 \text{ \AA}^{-1}$ (76 nm); by $t_w = 7863$ s the same bins give 0.01 to 0.13 (Figure 3c), so the complementary arrested fraction $1 - f$ has gone from 0.58 and 0.29 at those two Q values to 0.99 and 0.87 over approximately 47 min. As t_w grows, an increasing fraction of the scattering ensemble is therefore arrested within the accessible delay range, and this holds at every length scale probed. At every analyzed time f also increases monotonically with Q , end to end across the five bins by 9.0 standard deviations or more (Figure 3c and SI Section 5.3), so arrest develops first and most strongly at the largest length scale probed, 167 nm, and progressively less so toward 76 nm. That ordering holds separately within each elapsed-time group, so it reflects length scale rather than the overall trend with t_w , and it ties the growing low- Q SAXS structure to the arrest fraction measured on the very same frames. Note that XPCS does not locate those dynamically arrested regions, and $1 - f$ is not a direct geometric volume fraction, because the scattering contribution is weighted by structure and contrast. The coarsening signature is nonetheless clear: the largest structural features become the least mobile.

The fast mode loses amplitude without losing speed. $\langle\tau_{\text{fast}}\rangle$ decreases with Q at the four elapsed times where the fast mode is resolved, with a near-diffusive exponent of -2.08 to -2.37 , so the collective diffusion coefficient $\langle D \rangle = 1/(\langle\tau_{\text{fast}}\rangle Q^2)$ is a meaningful single number at each time. It moves non-monotonically between 1.7 and $5.5 \times 10^7 \text{ \AA}^2 \text{ s}^{-1}$, a factor of three

178 end to end, with no systematic trend (Figure 3c). Across the same interval f falls by a factor
 179 of five at the smallest length scale and to zero at the largest. The amplitude therefore carries
 180 the systematic change and the rate does not. A thermoreversible nanoparticle gel shows
 181 the same pairing of a plateauing diffusive fast time scale with a declining fast amplitude.²²
 182 Both stretching exponents stay below unity at every elapsed time, the signature of a broad
 183 distribution of relaxation times rather than a single process; the product of p_{fast} with the
 184 Q exponent is 1.3 to 1.5 rather than the value of 2 that subdiffusive motion of identical
 185 scatterers would give.³⁵ The fast amplitude and the fast relaxation time are not equally
 186 well determined at the last elapsed time: f is set by the height of the plateau and remains
 187 measured, whereas $\langle\tau_{\text{fast}}\rangle$ needs a resolved decay and is lost once f falls to a few percent,
 188 so the lowest- Q bin is dropped from the exponent there. τ_{slow} is not used for mechanistic
 189 scaling because it lies beyond the acquisition window (Figure S8). Correlation functions at
 190 the remaining Q values are shown in Figure S9. Using fresh material for every acquisition is
 191 a precaution against radiation damage: it keeps cumulative dose from tracking t_w , so the
 192 trend in f cannot be a dose artifact. A separate control at 23–41 times the flux used here
 193 shows no systematic change in g_2 with dose rate (Figure S10 and SI Section 6).

194 The growth of low- Q SAXS intensity and the decline of the fast fraction are two com-
 195plementary aspects of network formation. This coupling is consistent with network-forming
 196 arrested phase separation in a dynamically asymmetric protein–solvent mixture, for which
 197 concentration fluctuations produce a connected dense phase and viscoelastic constraints
 198 suppress relaxation.^{24,25,27,36} Related ELPs form bicontinuous networks with gel-like elasticity
 199 and stress-relaxation times well beyond 10^3 s.^{20,21} In associative networks built from the same
 200 pentapeptide family, the exchange kinetics of the associating blocks set the network relaxation
 201 time.³⁷ The present measurements add a local structural element from a separate experiment:
 202 at high temperature (VPAVG)₃₀ carries thermally reversible β -sheet-like contacts. As the
 203 low- Q assembly grows, it loses mobility fastest at the lowest Q , so the largest structures
 204 are the first to stop moving. That is the ordering a coarsening network should show. The

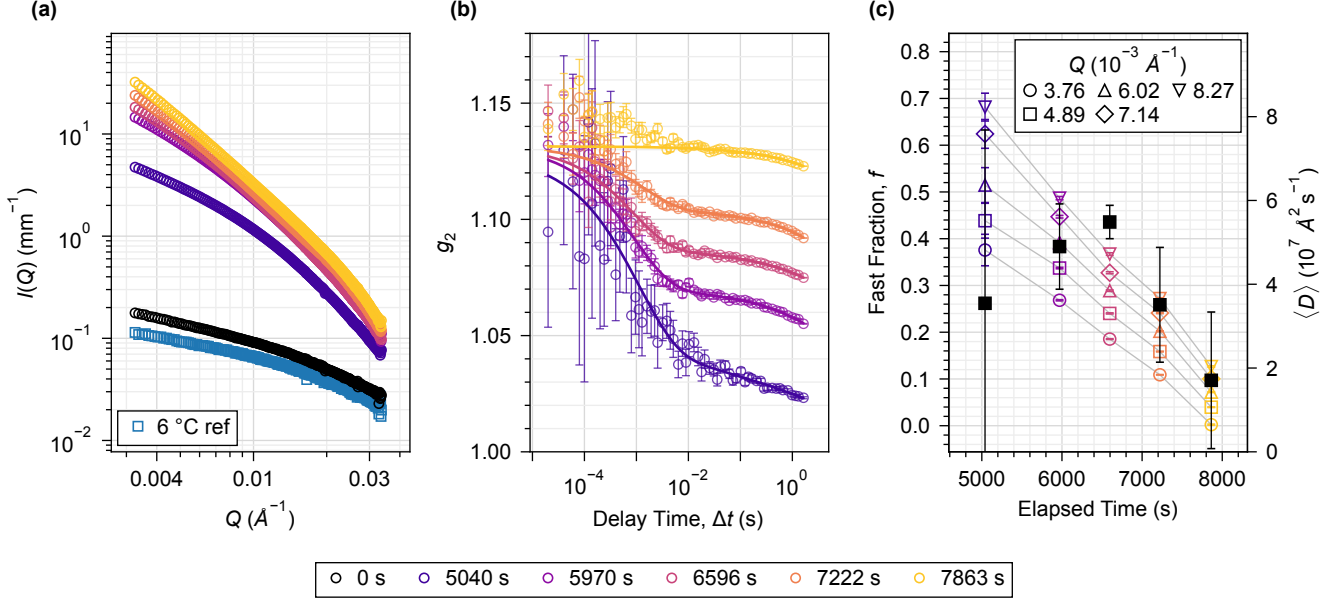


Figure 3: Isothermal structure evolution and dynamic arrest of (VPAVG)₃₀ at 30 °C. (a) Buffer-subtracted SAXS on an absolute scale (absolute scattering cross section per unit volume, calibrated as in Figure S6), showing continued growth of low- Q density contrast. Panel (a) alone also carries the 6 °C reference (blue squares, keyed separately) and the 0 s profile; (b) and (c) cover the five later times only. Full series in Figure S5. (b) Representative $g_2(Q, \Delta t)$ at $Q = 0.00376$ Å⁻¹; solid lines are global two-mode fits, and the error bars are the measured pointwise g_2 uncertainties. With elapsed time, the amplitude of the first decay decreases and the persistent plateau rises. Correlation functions and fits at the four other Q values are given in Figure S9. (c) Left axis, open coloured markers: the fitted fraction of fast dynamics, f , decreases at all five Q values; grey lines connect points at the same Q and are guides to the eye. Right axis, filled black squares: the mean collective diffusion coefficient of the fast mode, $\langle D \rangle = 1/(\langle \tau_{\text{fast}} \rangle Q^2)$, averaged over the Q bins where the fast mode is measured. The amplitude falls monotonically while the rate does not. The two outer times carry large uncertainties because the two-step decay is only just resolved at 5040 s and f has nearly vanished at 7863 s; those points are shown rather than dropped. Error bars are one-standard-deviation fit uncertainties for f and the spread across Q bins for $\langle D \rangle$. Relaxation-time scaling is given in Figure S8. Colors denote elapsed time throughout, keyed along the bottom; each elapsed time labels a group of averaged acquisitions as defined in the text.

separate experiments cannot place the ordered contacts at the junctions. They are nonetheless compatible in mechanism: arrest by discrete local conversion is the signature cooperative junction formation would leave, since a region locks when enough contacts form together rather than stiffening as each one tightens. β -sheet assembly is nucleated,³⁸ self-amplifying once initiated,³⁹ and grows by templated addition onto a self-complementary interface.⁶

The route to arrest matters for what a partially arrested network can do. A medium that stiffens throughout retards every solute within it. One that converts region by region keeps mobile pockets until the arrested phase percolates, so transport should follow the connectivity of those pockets rather than a bulk viscosity, and release should hold and then fall at a threshold rather than tail off. Because the arrested fraction evolves in time and can be measured, a network interrupted at a chosen elapsed time carries a defined fraction, a handle that composition alone does not give. The same question is open for biomolecular condensates, whose maturation from liquid to solid governs their function and their role in aggregation.^{40,41} If maturation proceeds the same way, a partially matured condensate keeps interior regions that still exchange, and the liquid-to-solid transition is a connectivity threshold rather than a gradual drift. Condensates are too small and too dilute for the measurement reported here. A concentrated, sequence-defined mimic is not, and in this one arrest is local.

Four limitations bound the interpretation. First, the β -sheet-like assignment rests on WAXS peak positions rather than spectroscopy; attenuated-total-reflection infrared through a thermal cycle is the natural orthogonal test, circular dichroism being poorly suited to a sample this turbid (Figure 1). Second, the high- Q ordering and low- Q dynamics come from separate experiments, so no time-resolved causal relation between peak growth and arrest is established. Third, the 2×10^{-5} –1.6 s delay range quantifies the fraction arrested on that time scale rather than the infinite-time nonergodicity factor; linear shear rheology reaches the terminal response directly, and pairing the two on a preparation large enough to charge a rheometer is the natural next step. Fourth, the isothermal series comes from one

loaded sample, and the quoted uncertainties are fit uncertainties rather than between-sample variability, so the trend in f is a single-preparation observation. Subject to those constraints, the combined data support network-forming arrested phase separation containing candidate, thermally labile β -sheet-like junctions.

In summary, SA-XPCS resolves how a sequence-defined protein polymer arrests. The amplitude of the mobile population falls by a factor of five at the smallest length scale and to zero at the largest, while its near-diffusive relaxation rate moves by a factor of three with no systematic trend. Arrest therefore proceeds by converting material out of that population rather than by slowing it, and it reaches the largest length scales first. SAXS/WAXS on the same construct resolves a reversible transition to a hierarchical assembly carrying β -sheet-like local order that vanishes on cooling. A protein network can therefore arrest region by region while the contacts that build it remain thermally erasable.

Experimental Section. The Sanger-verified construct encoded an N-terminal His-tagged (VPAVG)₃₀ protein (complete sequence in the Supporting Information). Recombinant (VPAVG)₃₀ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L⁻¹ HEPES, 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (Figure S1b). Protein concentration was determined from A_{280} using the sequence-specific extinction coefficient ($\epsilon_{280} = 1490 \text{ M}^{-1} \text{ cm}^{-1}$, 15,461 Da construct); the final concentration was approximately 100 mg mL⁻¹.

Temperature-dependent SAXS/WAXS measurements were performed at APS 12-ID-B with a 13.3 keV beam of approximately $100 \times 150 \mu\text{m}^2$ and an incident flux of approximately 5×10^{11} photons s⁻¹. At each set point the capillary was held at temperature for at least 10 min before exposure. SAXS and WAXS were acquired simultaneously in 0.1 s exposures with an Eiger2 9M at approximately 2 m and a Pilatus 300K at 0.42 m. SA-XPCS was performed at APS 8-ID-I with a 10.94 keV, $10 \times 10 \mu\text{m}^2$ partially coherent beam and a Rigaku XSPA-500k detector 7.8 m downstream. Each acquisition contained 100,000 frames at 50 kHz

(20 μ s per frame, 2 s total). Further acquisition, fitting and radiation-control details are in the Supporting Information.

Supporting Information

Recombinant synthesis and purification; gene-construction sequences and SDS-PAGE analysis; sample environments; SAXS/WAXS calibration; SA-XPCS analysis and global fitting; speckle-contrast calibration; radiation controls; Guinier analysis; complete isothermal SAXS evolution; length-scale-dependent XPCS parameters; correlation functions at all measured Q values; thermal-cycling repeatability; a still frame and link for the video of the reversible thermal transition; code and reduced-data availability; and AI-assisted development (PDF).

Data and Code Availability

Reduced data and analysis/figure-generation code are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the 8-ID-I and 12-ID-B datasets used for the figures, radiation-control data, and analysis scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

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AI-assisted manuscript and code preparation. The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and with development of the data-analysis and figure-generation code. The Claude models were accessed through Argo, Argonne National Laboratory’s internal generative AI interface (see Supporting Information). That code and the reduced data it operates on are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Conflict of Interest

The authors declare no competing financial interest.

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